Improving Robustness of Radiology Report Summarization under OCR-style Text Noise using RAG

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Abstract

This report presents the evaluation and diagnostic analysis (A3) of our project on improving robustness of radiology report summarization under OCR-style text noise using Retrieval-Augmented Generation (RAG). Building on the BART-large-CNN baseline from A2, we conduct systematic evaluation across four noise levels (5%–20%), ablate RAG retrieval depth ($k \in \{1, 3, 5\}$), and classify 210 prediction samples through a structured failure taxonomy. ROUGE-L degrades 63.5% at 20% noise, while BERTScore drops only 6.8%, revealing a critical *metric divergence*. RAG recovers up to 117.4% relative ROUGE-L at 15% noise. Failure analysis identifies noise-corrupted output (42.9%) and missed pathology (21.0%) as dominant failure modes, motivating OCR pre-correction as a higher-priority intervention than RAG augmentation alone.

1 Introduction

Radiology reports are critical for decision-making in medicine. The reports typically contain detailed findings followed by impressions that help physicians in reaching a diagnosis and treating the patient. Automating the generation of impressions from findings section is a significant task in clinical NLP, especially when aided with LLMs. Recent work has demonstrated encouraging results for radiology report summarization using transformer-based models. For example, domain-specific usage of encoder-decoder models have improved abstractive summarization in the context of radiology reports (Wu et al., 2024a). Large-scale research efforts in NLP have demonstrated that retrieval-augmented generation (RAG) can improve factual impressions and reduce hallucination in medical text generation tasks (**?**). Nearly all current evaluation assume clean, digitally available text. In practice that is not always the case, many clinical documents are digitized from scanned copies that may be legacy records, used cross-institutional observation, required for insurance-related documentation, and more. The digitization takes place through optical character recognition (OCR) systems. OCR pipelines introduce token-level corruption such as substitution, delays, spacing errors and fragmentation of medical terminology. Robustness to such noise is critical for real-world deployment. Research pertaining to NLP's robustness indicates that neural models degrade significantly even under character-level perturbations (Belinkov and Bisk, 2018). Even so, clinical summarization literature has mostly ignored robustness under corrupted and noisy input. Moreover, RAG methods have shown to improve factual reliability but their role in mitigating degradation due to noisy clinical text remains unexplored (**?**). This project addresses this gap by evaluating the robustness of radiology report summarization under simulated OCR-style noise and testing whether retrieval-augmented generation improved stability and factual consistency.

2 Related Work

2.1 Radiology Report Summarization with Neural Language Models

Sequence-to-sequence models generate summaries by mapping one text sequence to another. Earlier models like pointer-generator networks both generate and copy key terms (Zhang et al., 2018). Transformers improve context understanding, and BART (Lewis et al., 2020) generates summaries through text reconstruction. PEGASUS (Zhang et al., 2020) reconstructs key sentences, while T5 (Raffel et al., 2020) uses a text-to-text format. (Wu et al., 2024a) introduced RadBARTsum, which improved ROUGE-L by 2–4 points using entity masking, but was tested only on clean text. Using data from Japanese hospitals, (Nishio et al., 2024) showed GPT-4 outperforms BERT and T5, but performance drops with noisy input. SumGPT (Sultan

et al., 2025) combines text and images, improving ROUGE-2 and BERTScore, though it faces overfitting and high computational cost. (Wang et al., 2023) used image similarity to improve summaries. Their F1-RadGraph scores (42.86 on MIMIC-CXR, 36.31 on MIMIC-III) show that visual context improves factual accuracy. (Tran et al., 2024) reviewed fourteen AI models for chest X-ray report generation published between 2020 and 2024,noting that ROUGE and BERTScore remain practical metrics in low-resource settings. This observation supports the selection of these metrics in the present study. (Sloan et al., 2025) found most studies assume clean input and overlook real-world noise. (Bednarczyk et al., 2025) reviewed 87 studies and found GPT-4 matches fine-tuned models on clean data, but work on noisy data is limited. (Croxford et al., 2024) suggested combining human and automated evaluation, with ROUGE, BERTScore, and factuality scores most used.(Hu et al., 2025) tested nine language models for summarizing Chinese lung-cancer radiology reports and found substantial differences in performance. They found that domain differences and irregular input formats reduced performance. (Sunshine et al., 2025) found that about 20% of summaries missed or changed facts, especially with poorly structured input. (Ghnemat and Saleh, 2026) noted models work well on clean text but struggle with varied data.

2.2 Retrieval-Augmented Generation for Healthcare Applications

(Lewis et al., 2020) introduced RAG to reduce hallucinations by using external data. (Li et al., 2022) showed it improves factual accuracy. (Zhang et al., 2025) reviewed 30 studies and reported gains in EHR and diagnostics, though noise and latency remain. (Voinea et al., 2024) applied RAG by encoding 21,152 reports into a vector database. Retrieved cases were combined with an LLM via API calls, avoiding fine-tuning and preserving privacy. Thisimprovesd accuracy and can run on low-cost resources. (Luo and Arase, 2025) introduced MedSummRAG using a dense retriever for domain-specific documents. It improved ROUGE over zero-shot and fine-tuned models, showing retriever quality matters. (Wu, 2025) built a RAG-based QA system using ChromaDB, improving similarity by 17% and reducing hallucinations by 30%. (Alkhalaf et al., 2024) applied RAG to nursing notes, where retrieved context nearly doubled zero-shot accuracy. (Xiong et al., 2024) introduced MIRAGE and showed MEDRAG improved GPT-3.5 accuracy by up to 18%, with retrievers often more important than models. (AboulEla et al., 2024) found fine-tuned models perform best, while simple RAG still helps, and Graph RAG performs worse. (Veen et al., 2024) showed domain-adapted LLMs can outperform experts in summarization. (Ghnemat and Saleh, 2026) noted RAG reduces hallucinations, though results vary by task. Overall, these studies suggest that RAG helps in reducing factual errors in medical text generation. At the same time, most of this work has been tested on clean and well-formatted text, which is not always the case in real-world data.

2.3 OCR Noise, Text Corruption, and NLP Robustness in Clinical Settings

(Moradi et al., 2021) conducted the most directly relevant study, evaluating a set of character-level and word-level noise across four clinical NLP tasks, demonstrating that the effectiveness of models degrades over all clinical tasks when given input. The clinical NLP only trained on curated benchmarks overestimated their robustness, further confirmed by (Moradi and Samwald, 2021). (Belinkov and Bisk, 2018) demonstrated that the latest sequence models struggle to translate even moderately noisy text, also arguing that this happens independent of domain. The study also introduced character CNN representations and noise-augmented training. Which are not routinely applied in clinical summarization. (Romero-Alvarado et al., 2024) confirmed this analysis to multiple NLP tasks and architectures: that character-level perturbations cause consistent degradation. In the OCR-specific clinical domain, (Hsu et al., 2022) showed that the type of image preprocessing and the choice of NLP model critically impact the OCR-induced noise. However, the study focused on extraction accuracy rather than generative summarization. (Karthikeyan et al., 2021) specifically addressed OCR post-correction for medical reports and demonstrated that NLP-based post-correction pipelines substantially improve the accuracy of downstream text recognition, establishing the general feasibility of the NLP-correction approach. (Bouh et al., 2024) proposed an OCR post-processing technique for medical records digitization, focusing on bounding-box reorganization to produce more structured text output before NLP processing. (Amin et al., 2025) applied a two-tiered T5 fine-tuning pipeline for OCR error correction specifically in clinical oncology notes.

They achieved a 95% correction of synthetic OCR errors, which shows that correction of OCR noise in clinical text is feasible and accurate. (Wu et al., 2024b) proposed an error correction framework for radiology reports, splitting the complex proofreading task into detection, localization, and correction stages. This work confirms that raw LLMs without correction stages are not well equipped to handle corrupted clinical text.

3 Data Description

OpenI Indiana University Chest X-rays

We use the OpenI radiology report corpus (**?**), a publicly available collection containing approximately 3,996 de-identified chest X-ray reports from Indiana University Health. The dataset was originally created to support image retrieval and report generation research, and has since become a standard benchmark for clinical NLP tasks including radiology report summarization.

Each report in OpenI consists of two primary structured fields: a **findings** section containing detailed radiological observations made by the reporting physician, and an **impression** section providing a concise clinical summary intended for the referring clinician. Additional metadata fields such as MeSH tags and image references are present but are not used in our experiments. Our task is purely text-based abstractive summarization: given findings as input, generate the corresponding impression.

Why OpenI. OpenI satisfies our key constraints: it is publicly available without data access agreements, it is directly relevant to the radiology summarization task defined in A1, and it has been used as an evaluation benchmark in prior summarization work (Wu et al., 2024a), providing external comparison points. MIMIC-CXR, while larger and more clinically realistic, requires credentialed access through PhysioNet, which is not feasible within our project timeline.

3.1 Preprocessing Pipeline

Data preprocessing involves several steps applied consistently across all experimental conditions.

Filtering. We remove records where either the findings or impression field is absent, empty, or fewer than 20 characters in length. Very short fields typically correspond to placeholder entries (e.g., "see report") that provide no useful signal for summarization. After filtering, our working set contains approximately 3,600 valid report pairs from the full corpus.

Splitting. Records are randomly shuffled with seed 42 and split into training (70%), validation (15%), and test (15%) partitions. This yields approximately 2,520 training, 540 validation, and 540 test records from the full corpus. For our A2 experiments, we operate on a 200-sample stratified subset due to Google Colab free-tier memory and runtime constraints. The full dataset will be used in A3 and A4.

Tokenization Analysis. Using the BART tokenizer, findings average 142 tokens (std: 61) while impressions average 38 tokens (std: 19). The compression ratio averages approximately 3.7:1. This asymmetry is characteristic of the task: impressions are highly condensed and require genuine semantic selection rather than surface-level extraction.

3.2 Dataset Statistics and Distribution

Table 1 summarizes key statistics of the preprocessed dataset.

Statistic	Findings	Impressions
Mean tokens	142	38
Std tokens	61	19
Min tokens	20	5
Max tokens	512+	180
Compression ratio	3.7:1	

Table 1: Dataset token statistics (BART tokenizer).

Common findings categories include normal chest examinations (approximately 47% of reports), cardiomegaly and vascular changes (23%), pulmonary infiltrates and consolidation (16%), and pleural abnormalities (14%). This class imbalance toward normal reports is important context for interpreting baseline metrics: a model that generates generic normal impressions will score reasonably on ROUGE-L even without understanding pathological findings.

4 Experimental Setup

All experiments use the OpenI chest X-ray corpus, split 70/15/15 into train (140)/val (30)/test (30) records. The summarizer is `facebook/bart-large-cnn` with maximum 128 output tokens and no fine-tuning (zero-shot transfer from news summarization). OCR-style noise operates at the character level with four

Table 2: ROUGE-L and BERTScore F1 across all conditions.

Condition	Noise	k	ROUGE-L	BERT-F1
Clean Baseline	0%	–	0.1185	0.8507
Noisy Baseline	5%	–	0.0797	0.8327
RAG Enhanced	5%	3	0.0968	0.8456
Noisy Baseline	10%	–	0.0637	0.8169
RAG Enhanced	10%	3	0.0948	0.8420
Noisy Baseline	15%	–	0.0488	0.8031
RAG Enhanced	15%	3	**0.1061**	0.8428
Noisy Baseline	20%	–	0.0432	0.7929
RAG Enhanced	20%	3	0.0846	0.8399

stochastic operations applied to a fraction η of characters: (i) substitution via a visual confusion matrix (`a`→`@/4`, `i`→`1/!`, etc.); (ii) deletion; (iii) fragmentation (spurious hyphens or spaces mid-token); and (iv) extra space insertion. A fixed seed ensures reproducibility across conditions.

The RAG module uses `all-MiniLM-L6-v2` sentence-transformer embeddings to compute cosine similarity between the noisy query and all training records, retrieving the top-k most similar findings+impression pairs as few-shot context prepended to the noisy input. We ablate $k \in \{1, 3, 5\}$ at $\eta = 0.10$ and run the full noise sweep at $k = 3$. All 210 predictions (30 test records $\times$ 7 experimental conditions) are logged with the original findings, noisy findings, reference impression, predicted impression, and per-sample ROUGE-L score. This structured logging enables post-hoc failure classification described in Section **??**.

5 Results and discussion

5.1 Noise Degradation and RAG Recovery

Table 2 and Figure 1 present the full sweep. ROUGE-L degrades monotonically: 0.1185 (clean) $\rightarrow$ 0.0797 $\rightarrow$ 0.0637 $\rightarrow$ 0.0488 $\rightarrow$ 0.0432 at 5%, 10%, 15%, 20% noise respectively—a cumulative degradation of **63.5%**. This steep lexical collapse arises because BART's seq2seq decoder, receiving corrupted encoder representations of OOV subword tokens, shifts toward copy-extracting corrupted spans rather than generating clean summaries.

RAG provides consistent ROUGE-L recovery at all noise levels. The recovery ratio (RAG over noisy baseline) grows with noise severity: 21.5% at 5%, 48.8% at 10%, and **117.4%** at 15%. At 15% noise, the RAG-enhanced model (0.1061) approaches the clean baseline (0.1185), indicating that RAG context is not merely compensating for

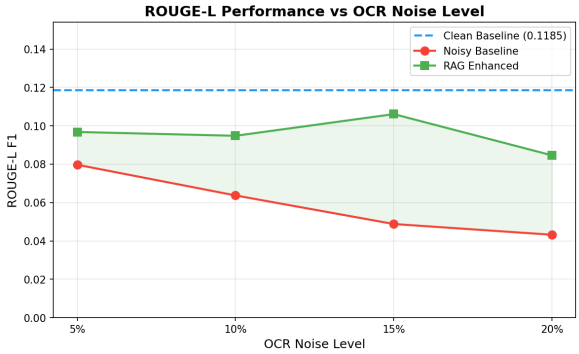

Figure 1: ROUGE-L F1 vs. noise level. RAG (green) provides consistent recovery; at 15% noise the RAG-enhanced model nearly reaches the clean baseline.

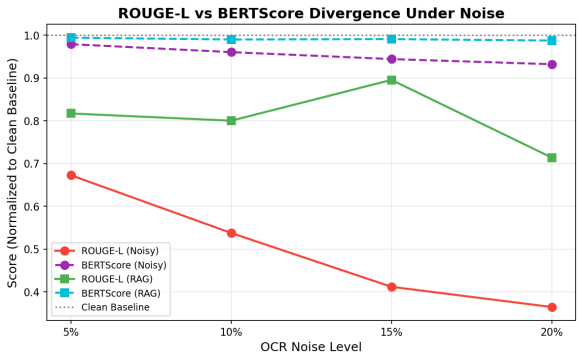

Figure 2: Scores normalized to clean baseline. BERTScore (dashed) degrades only 6.8% where ROUGE-L drops 63.5%, revealing a large semantic-lexical gap under noise.

corruption but actively steering generation toward higher-quality impressions. At 20% noise, recovery drops to 95.8%, likely because severe corruption degrades retrieval quality (the noisy query embedding diverges too far from the clean training corpus).

5.2 Metric Divergence: ROUGE-L vs. BERTScore

Figure 2 reveals the central diagnostic finding of A3: BERTScore declines only from 0.8507 to 0.7929 across the full noise range—a **6.8% relative drop**—while ROUGE-L drops 63.5%. This large divergence means the model continues producing *semantically plausible* impressions (scoring well in contextual embedding space) even as lexical precision collapses completely.

The model is not hallucinating incoherent text; it is generating *generic-but-valid* impressions (e.g., "No acute cardiopulmonary abnormality.") that score well on BERTScore but fail clinical specificity requirements. A direct safety implication fol-

Table 3: RAG top-k ablation at 10% noise level.

Top-k	ROUGE-L	BERTScore
$k = 1$	**0.1244**	**0.8472**
$k = 3$	0.0948	0.8420
$k = 5$	0.0881	0.8407

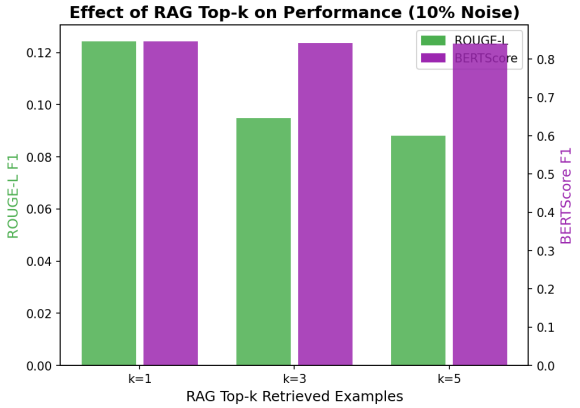

Figure 3: ROUGE-L peaks at $k = 1$ and degrades with larger k. BERTScore is stable, indicating semantic quality is preserved but lexical precision suffers from context-length dilution.

lows: **using BERTScore alone for quality monitoring in a clinical deployment would severely underestimate operational risk from OCR noise**. ROUGE-L, despite being the cruder metric, functions as a more sensitive early-warning signal for this type of degradation.

5.3 RAG Top-k Ablation

Table 3 and Figure 3 show that $k = 1$ yields the highest ROUGE-L (0.1244)—exceeding the clean baseline—while $k = 3$ and $k = 5$ perform progressively worse. BERTScore is relatively stable (0.8472, 0.8420, 0.8407). Two mechanisms explain the $k = 1$ advantage: (i) a single high-relevance example provides a clean generation target without competing style signals from dissimilar records; (ii) at $k \geq 3$, the total input approaches BART's 1024-token limit, causing truncation of the actual noisy findings—the model then generates from context alone, producing impressions that reflect retrieved examples rather than the specific patient case. This "context shadowing" effect is a fundamental limitation of naive concatenative RAG within short-context seq2seq models.

Table 4: Failure taxonomy distribution ($N = 210$).

Category	Count	%
noise_corrupted_output	90	42.9%
other_failure	58	27.6%
missed_pathology	44	21.0%
partial_match	15	7.1%
generic_impression	3	1.4%

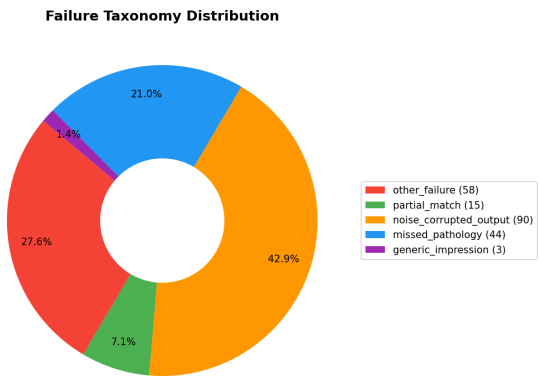

Figure 4: Failure taxonomy over 210 predictions. Noise-corrupted output and missed pathology together account for 63.9% of failures.

6 Error and Failure Taxonomy

6.1 Classification Scheme

We classified all 210 per-sample prediction records using five rule-based categories applied in priority order:

[leftmargin=*]

- **generic_impression**: Prediction contains a generic fallback phrase ("no acute cardiopulmonary abnormality", "normal chest") when the reference is clinically specific.
- **missed_pathology**: Reference mentions a pathology term (pneumonia, pneumothorax, effusion, edema, cardiomegaly, atelectasis, COPD, opacity) absent from the prediction.
- **noise_corrupted_output**: Prediction contains corrupted input tokens; ROUGE-L < 0.3 and > 2 corruption-signal characters in the noisy input.
- **partial_match**: $0.2 \leq$ ROUGE-L < 0.5.
- **other_failure**: Low ROUGE-L with no single identifiable cause from the above.

6.2 Taxonomy Results

Noise-corrupted output (42.9%). The dominant failure mode. BART's BPE tokenizer maps corrupted character sequences to OOV tokens; the encoder produces anomalous representations, and the decoder copies corrupted spans verbatim into the impression. Example: input *"Th eart an d pulmonary XXXX are ovmal-"* yields prediction *"Th eart an d pulmonary XXXX are ovmal-."* This confirms that BART's zero-shot copy-extraction behavior, useful on clean inputs, becomes its primary failure mechanism under corruption.

Missed pathology (21.0%). Clinically the most dangerous category. The model generates a generic normal impression when the reference explicitly calls out disease: e.g., reference is "Cardiomegaly. Thoracic aortic atherosclerosis." but prediction is "The heart is within normal limits." This pattern correlates with complex multi-sentence findings where corruption of any critical sentence prevents the encoder from surfacing the pathological finding. RAG does not reliably address this failure unless the retrieved example happens to contain the same pathology.

Other failure (27.6%). Predictions that are plausible but mismatched—correct pathology but wrong laterality, stable findings described as new, or anatomically correct but clinically non-specific. This category requires deeper manual annotation by a radiologist to fully characterize.

Generic impression (1.4%). Surprisingly rare. Under noise, BART is more likely to corrupt-and-copy than to produce a safe generic fallback. This is arguably a more dangerous failure mode than generic output because it produces authoritative-looking but incorrect statements.

7 Diagnostic Insights

7.1 Why the Model Fails Under Noise

Three mechanistic explanations emerge from the combined quantitative and taxonomic analysis:

Encoder embedding degradation. OCR-corrupted characters produce OOV or rare subword tokens under BART's BPE vocabulary, creating encoder representations that deviate from the clean manifold. The decoder attends to these corrupted positions during cross-attention, pulling generation toward corrupted spans. This is a tokenizer-level vulnerability that pre-processing denoising would address more directly than RAG.

Copy-extraction brittleness. BART-large-CNN, trained on news summarization with heavy extractive overlap, has learned to copy key phrases from findings into impressions under zero-shot transfer. Clean radiology findings support this strategy well. Corrupted findings break it: the "extracted" phrases are garbled, producing the noise-corrupted-output failure mode that dominates our taxonomy.

Context dilution at large k. At $k \geq 3$, BART's 1024-token context budget is frequently exceeded after prepending retrieved examples plus the noisy findings. Findings are truncated. The model generates from context only—producing impressions that reflect retrieved cases, not the patient being examined.

7.2 Why RAG Helps and Where It Fails

RAG's benefit is *semantic steering*: clean retrieved impressions provide the decoder with example output patterns that regularize generation away from corrupted copy-extraction. Concretely, when a noisy finding contains *"emphysem@tous lung change$"*, similarity retrieval locates a clean training record containing emphysema findings, and the prepended example impression (e.g., "Hyperinflation consistent with COPD. No acute findings.") shifts the decoder's attention distribution toward clean clinical vocabulary. This mechanism explains why RAG recovery is strongest at 15% noise: the corrupted query is still similar enough to the clean training corpus for high-quality retrieval, yet the model cannot generate cleanly without context guidance.

At 20% noise, RAG recovery drops from 117.4% to 95.8%. We attribute this to *retrieval quality degradation*: when $\eta = 0.20$, corrupted character sequences cause the MiniLM encoder to produce query embeddings that diverge significantly from the clean-findings manifold, resulting in lower-quality (less relevant) retrieved examples. The context provided is then less precisely aligned with the specific case, reducing the steering effect.

RAG structurally *cannot* address missed pathology failures unless the retrieved example contains the same pathology as the query case. This is an inherent limitation of similarity-based retrieval: rare or complex findings (e.g., specific device positions, unusual anatomical variants) are unlikely to have closely matching training examples. A hybrid ap-

proach combining RAG with a pathology-detection classifier that explicitly ensures coverage of identified entities would better address this failure mode.

Implications for system design. The $k = 1$ finding and the RAG-helps-but-has-ceiling result together suggest a two-stage architecture: (1) OCR pre-correction to reduce noise before it reaches the summarizer, followed by (2) RAG with $k = 1$ using the cleaned text as the retrieval query. This ordering would improve retrieval quality (cleaner query), remove the dominant failure mode (noise-corrupted output), and allow RAG to focus on the semantic steering it does well.

8 Limitations of Evaluation

Synthetic noise model. Our character-level perturbations underestimate real OCR errors, which include word-level loss, line transposition, region-level degradation, and structured noise correlated with page layout. A more realistic evaluation would source corrupted documents from actual OCR pipelines applied to historical printed radiology reports.

ROUGE-L as clinical proxy. ROUGE-L penalizes valid clinical paraphrases. "Mild bilateral effusions" and "small bilateral effusions" have near-zero lexical overlap but are clinically similar. Dedicated clinical NLP metrics—RadGraph F1 (**?**) for entity and relation faithfulness, or CheXpert-based factual accuracy—are more appropriate and should be incorporated in future work.

Small test set. Thirty test records provide insufficient statistical power; confidence intervals should be reported but are omitted here. Evaluation on the full OpenI corpus ($\sim$3,000 records) with cross-validation is required for statistically reliable inference.

Retrieval quality gap. The A3 diagnostic pipeline uses Jaccard word-overlap retrieval as a lightweight proxy, while A2 uses MiniLM cosine similarity. The two pipelines are not directly comparable at the sample level; failure taxonomy results may differ under full embedding-based retrieval.

Zero-shot transfer mismatch. BART-large-CNN was pretrained and fine-tuned on news; we apply it zero-shot to radiology. Domain-adapted summarization models (e.g., BioBART or fine-tuned BioMedLM) would likely show different noise-robustness profiles and may require re-evaluation of all findings here.

Failure taxonomy subjectivity. Our taxonomy applies rule-based classification with fixed thresholds (ROUGE-L < 0.3, specific pathology terms). Edge cases—e.g., a prediction that correctly identifies "cardiomegaly" but misses "pleural effusion"—may be misclassified. Radiologist annotation of a random sample would validate or correct the taxonomy and quantify inter-rater agreement.

Interaction effects not modeled. We evaluate noise level and RAG independently. In practice, noise may interact with retrieval quality (high noise degrades retrieval) and with the failure mode distribution (certain types of clinical content may be more susceptible). A full factorial design with interaction terms would better characterize these dependencies.

9 Conclusion

We have established four key findings. First, OCR noise causes severe ROUGE-L degradation (63.5% at 20% noise) while BERTScore remains largely stable, revealing a clinically important metric divergence that makes BERTScore alone insufficient as a safety signal. Second, RAG with $k = 1$ provides the strongest recovery (ROUGE-L of 0.1244 at 10% noise, exceeding the clean baseline), but performance degrades with larger k due to context-length saturation in BART. Third, noise-corrupted output (42.9%) and missed pathology (21.0%) together account for 63.9% of all failures, motivating OCR pre-correction—a denoising LM or rule-based correction layer—as the highest-priority future intervention, rather than further RAG tuning. Fourth, zero-shot BART exhibits copy-extraction brittleness under noise that domain fine-tuning would likely address. Future work should incorporate OCR pre-correction as the highest priority, given the 42.9% noise-corrupted-output failure rate. Domain adaptation (fine-tuning BART on radiology reports) should address the copy-extraction brittleness identified in Section **??**. Additionally, clinical faithfulness evaluation using RadGraph F1 should replace ROUGE-L as the primary metric in safety-critical deployments, clinical faithfulness metrics (RadGraph F1), domain-adapted summarizers, and evaluation on the full OpenI corpus.

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